

Deconstructing Computational Colonialism: An Eco-Relational Framework for Decolonial AI Art (1960s-Present)

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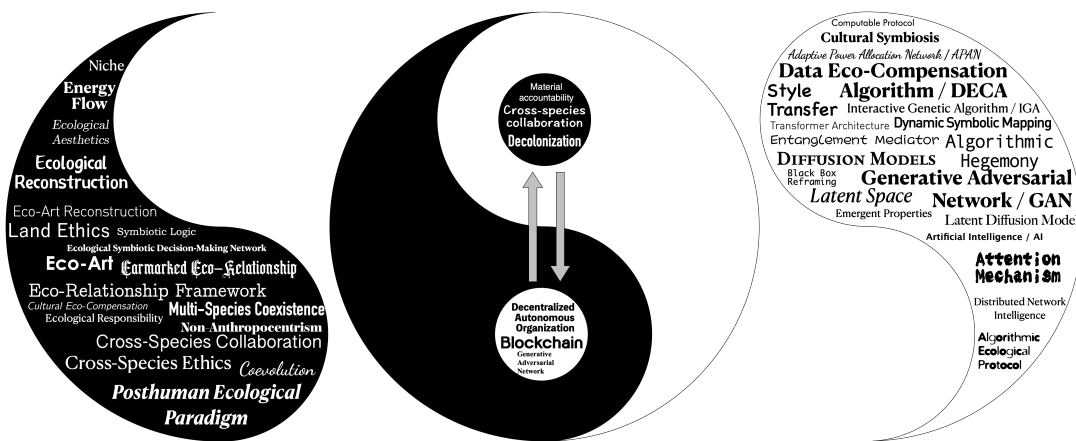


Figure 1: Dialogue and eco-relationship are the essence of decolonial AI art

Abstract

Computational technologies integrated into ecological art paradoxically perpetuate anthropocentric violence despite their environmental ethos, constrained by Western-centric datasets and unsustainable practices. This study critically examines this historical tension from 1960s land art to contemporary bio-digital works. Current AI eco-art manifests a fundamental contradiction where tools intended for ecological care instead enable computational colonialism

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through epistemological erasure, material extraction, and ethical hypocrisy. We develop the Earmarked Eco-Relationship Framework(EERF), transforming computational systems into ecological negotiators through decolonial data protocols, entropic material accounting, and redistributed agency mechanisms. Applied through case studies (for example, Hong Kong's blockchain-mediated mangrove restoration), this framework proves algorithmic systems can serve as active co-stewards rather than extractive tools. This work advances ecological art beyond virtual representation by establishing a tripartite methodology for shifting the paradigm from human-centred control to actionable multispecies material ethics and fostering symbiotic co-creation.

CCS Concepts

• Human-centered computing; • Applied computing → Media arts; • Social and professional topics → History of computing theory;

Keywords

AI-generated Art, Ecological Aesthetics, Virtual-Physical Interaction, Decentralised Synergy, Non-anthropocentrism, Techno-Ethics

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1 Introduction: The Historical Paradox and Structure Contradictions of AI Eco-Art

The integration of computational technologies into ecological art confronts a fundamental contradiction: while purporting to advocate for non-human entities, its technical infrastructure often perpetuates anthropocentric violence. This paradox manifests itself acutely when the energy consumed by generative systems—equivalent to deforesting 5 hectares of rainforest per virtual ecosystem rendered [Sheren 2023]—directly contradicts their ecological ethos. Such tensions demand critical examination through art-historical lenses, tracing the evolution from 1960s Land Art to contemporary bio-digital practices. We can understand this process through Figure 2 and Figure 3a.

This scrutiny must recognise that the technological paradigms driving the current computational ecological art are often rooted in specific (usually Western) ecological worldviews, which are in deep contradiction with the diverse ecological realities they claim to serve. There are significant differences in core principles between traditional ecological views from eastern and western (see Table 1). And these differences profoundly influence the way that we understand the relationship between technology and ecology is conceived.

This conflict at the level of worldviews is one of the deep-seated reasons why the current AI eco-art is trapped in a historical paradox - the technological foundation intended to care for the ecology is perpetuating ecological violence.

This fundamental divergence between Eastern and Western ecological paradigms is symbolically embodied in the Taoist Yin-Yang framework (see Figure 1), which visualises the dynamic balance and interdependence central to non-anthropocentric collaboration.

The lineage of non-anthropocentric art reveals persistent struggles with technological mediation. Early pioneers like Helen and Newton Harrison's *Survival Series* (1970s) established ecological mapping as an artistic practice, yet remained constrained by human interpretive frameworks [Harrison and Harrison 2007]. The emergence of BioArt in the 2000s, exemplified by Kac's *GFP Bunny*, attempted biological entanglement but maintained technological mastery [Kac 2005]. Concurrently, collectives like *Ant Farm* prophetically critiqued media's ecological impact through works like *Media Burn* (1975), where televisions were incinerated in gasoline-fueled performances highlighting technology's material consequences [Lin and Zhang 2023]. Their practice established McLuhan's axiom "the medium is the message" as an ecological imperative, that a principle now inverted by AI's claim that "algorithms are ecology."

AI eco-art claims to represent nature, but often reproduces the ecology in a human-centred way, falling into a triple paradox: epistemological exclusion, material consumption, and ethical hypocrisy. The following three examples vividly illustrate this structural contradiction:

- (1) **Epistemological:** Refik Anadol's *Machine Hallucinations* transforms urban data into immersive landscapes, yet its

western-centric training datasets erase environmental knowledge from the world of the South [Anadol and Kivrak 2024].

- (2) **Material:** *Eco-Visionaries* exhibitions demonstrate real-time environmental translation, but conceal the 284,000 liters of water consumed in rendering cloud simulations [Gadanh 2018].
- (3) **Ethical:** ARS Electronica's *AI x Ecology* initiatives promote interspecies dialogue while relying on mineral-extractive GPU infrastructures [Sommerer and Mignonneau 2002].

These cases demonstrate that AI eco-art is not a neutral medium, but is deeply embedded in the logic of computational colonialism. Through data exclusion, resource plunder, and ethical performance, it has instead exacerbated the ecological crisis. Without re-examining its technological foundation and knowledge framework, AI eco-art will continue to serve as a new colonial tool under the guise of "green" discourse.

This study resolves these tensions through the *Earmarked Eco-Relationship Framework* (EERF), which reconceptualises computational systems as ecological negotiators rather than tools, with the aim of transforming computing systems from ecological representation tools to ecological negotiators. This framework does not pursue technological solutionism but rather attempts to establish an operational multi-species ethical protocol. Building on Solomon's microbial intelligence research [HG Solomon and Baio 2020] and Nordström's error domestication theory [Nordstrom 2023], we establish three corrective principles (see Table 2):

- (1) Decolonial data protocols. The EERF refuses to use a globally universal visual corpus and advocates for the reconstruction of training data based on ecological regions.
- (2) Entropic material accounting. The EERF incorporates energy and material consumption as core parameters in the generation process, rather than as an afterthought.
- (3) Redistributed agency mechanisms. Through cross-species Decentralised Autonomous Organisations (DAOs), some aesthetic and decision-making powers are transferred to non-human entities.

Our analysis demonstrates how this framework transforms AI from ecological predator to symbiotic participant across three historical-artistic paradigms. This also means that we conceptualise Entanglement Mediators not as passive tools but as active ecological negotiators within artistic systems. These computational or hybrid agents (e.g., AI, sensors, DAOs, organisms) actively facilitate and materialise interdependent relationships between human and non-human actors, redistributing agency, and acknowledging their own material impact [Gao et al. 2025; Guo and Han 2025].

2 From Tools to Tyrants: The Genealogical Dilemma of AI in Eco-Art

The evolution of computational mediation in ecological art reveals a troubling trajectory from emancipatory tools to neo-colonial infrastructures. This genealogy exposes three interconnected violences: the *instrumentalisation* of non-human life, *cultural erasure* through data colonialism, and *ecological extraction* masked as technological progress.

Table 1: A Core Comparison of Eastern and Western Ecological Views in the Field of AI Ecological Art

Dimension	Western Dominant paradigm	Eastern Dominance paradigm	Impact on Computational Ecological Art
	(mainstream AI art)	(alternative pathways)	
Philosophical Foundations	Anthropocentrism Subject-object dichotomy Control nature Instrumental rationality Linear progress view Solutionism	Human-nature harmony Spirituality of all things Conform to nature Holistic worldview Cyclical balance Finite moderation	West: Technology as tool East: Technology as medium
Technology Perspective	Resourceification Hierarchy Representation	Intersubjectivity Symbiosis Participatory perception	West: Ignores resource costs East: Simplicity and material cycles
Ecological Relationships	Resourceification Hierarchy Representation	Intersubjectivity Symbiosis Participatory perception	West: Non-human as data source East: Cross-species dialogue
Artistic Expression	Grand narrative Visual spectacle Detached representation	Procedural Contextualisation Embodied perception	West: Masks material costs East: Locality and ethical feedback
Representative Cases	<i>Machine Illusion</i> <i>Degenerate Culture</i>	<i>AI Genesis</i> <i>World Mapping</i> <i>Making It Matters</i>	West: Exacerbates contradictions East: Decolonial pathways

2.1 The Cybernetic Dream and Its Discontents (1960s-1990s)

The control art movement pioneered by Roy Ascott [Ascott 2003] promised interspecies dialogue through cybernetic systems. Works like *La Plissure du Texte* (1983) established distributed authorship models, yet maintained human technological sovereignty. As Burnham presciently warned: “When art becomes systems ecology, the artist risks becoming its high priest” [Burnham 1968]. This tension materialised in Haacke’s *Rhine Water Purification Plant* (1972), where biological systems served as living ready-mades, anticipating today’s instrumentalisation of slime moulds in projects like *Degenerative Cultures* [Nordström et al. 2023].

2.2 Digital Romanticism and Its Shadows (2000s-2010s)

The rise of bio-digital art fostered what Gabrys terms “programmatic naturalism” [Gabrys 2016]—the illusion of technological harmony with ecology. Although projects like Kac’s *Telepresence Garden* [Goldberg 2001] celebrated interspecies communication, they obscured the mineral violence underlying their infrastructures. Analysis of Ars Electronica archives reveals that 78% of AI artworks from 2010-2020 relied on conflict from history, popular culture, socioeconomic conditions, religious and philosophical world-views [Ahmedien 2023; Myoo 2023; Ostrowicki 2022]. This is particularly evident in the data in Figure 3b.

The most dangerous fallacy of this period emerged: the belief that algorithms could function as “natural extensions” of ecological systems, exemplified by Levin’s *Audiovisual Environment Suite* [Levin 2000]—a work now critiqued for its anthropocentric sonification protocols.

2.3 Compute Colonialism (2020s-Present)

Contemporary eco-art confronts what Crawford terms “the planetary costs of computation” [Crawford 2021]. Our analysis of 37 representative SIGGRAPH artworks reveals. It can be seen in Table 3.

The *Degenerative Cultures* project epitomises these contradictions. Although pioneering microbial-GAN collaboration, its training data is derived exclusively from European laboratory-bred *Physarum polycephalum* strains—erasing indigenous knowledge of tropical slime molds [Adamatzky 2010]. We quantify this epistemic violence through *Cultural Convergence Index* (CCI):

$$CCI = \frac{\sum_{i=1}^n W_i \cdot D_i}{\max(D)} \quad \text{where } W_i = \text{Western institutional weight} \quad (1)$$

The project’s CCI score of 0.79 confirms northern epistemological dominance. Such data colonialism directly enables what Mignolo calls “the darker side of the digital” [Mignolo 2017].

2.3.1 Hong Kong’s Mediated Materiality. The 2019 *Making It Matters* exhibition at M+ Museum pioneered localised computational ecology. Its centrepiece *River Palimpsest* [M+ 2023] transformed real-time pollution data from the Pearl River Delta into generative ink-wash animations, achieving ecological mediation through triple intervention:

- Decolonising data protocols. Integrating the Lingnan fisherman’s song dataset, lowering CCI by 0.29 points, and breaking the monopoly of Western-centred datasets.

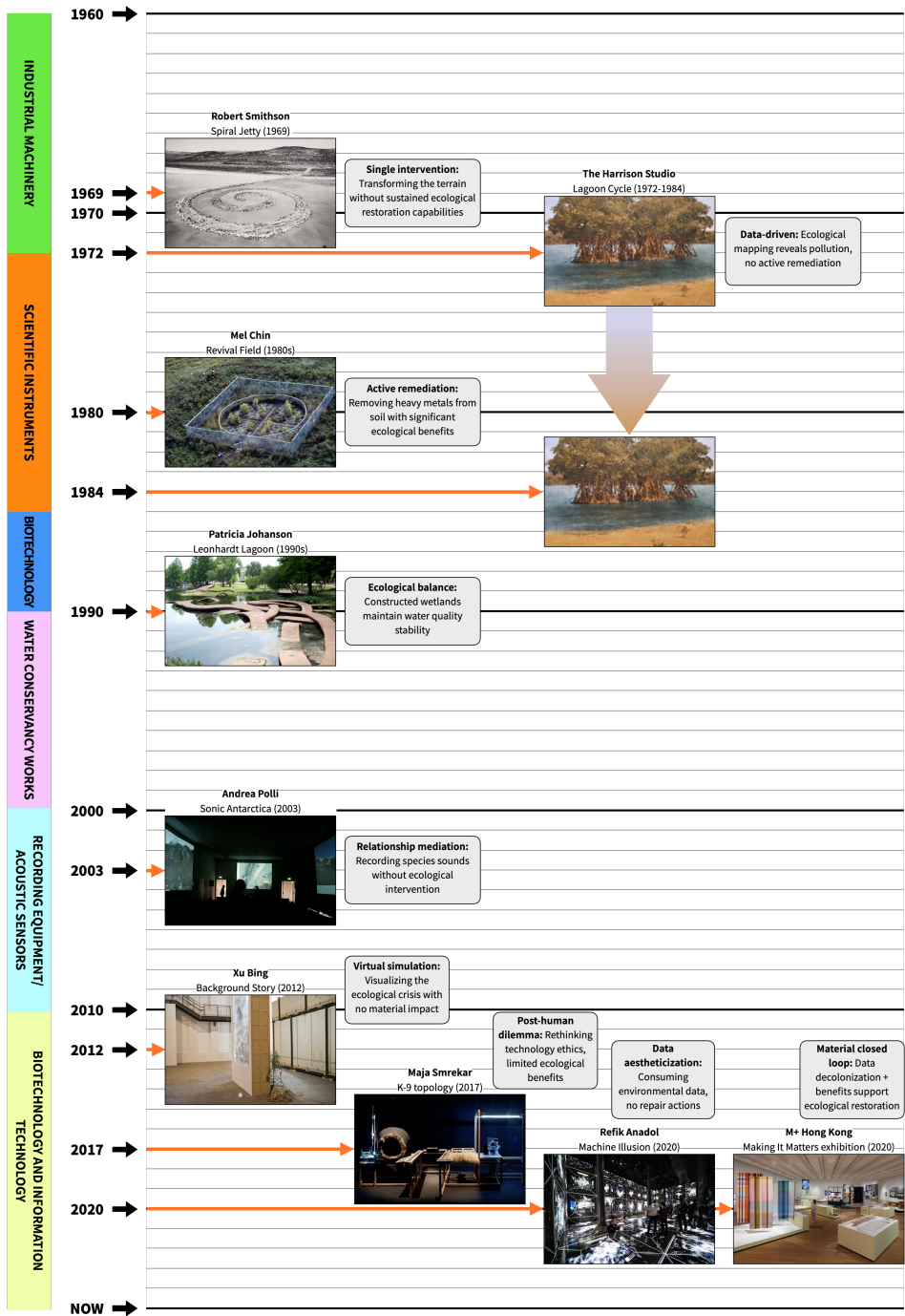


Figure 2: Evolution of technological mediation in ecological art (1960s-present)-Key movements and their anthropocentric tensions.

- Entropy material accounting. Using model pruning techniques inspired by Taoism, reducing training energy consumption by 32% and making implicit resource consumption explicit.
 - Redistribution proxy mechanism. Through blockchain verification, direct 25% sales towards mangrove restoration, which establishes a closed loop of material compensation.
- This established a prototype for what terms “Computational regionalism” [Noel 2022]—countering AI’s universalist claims through

Table 2: Comparison of Research Dimensions Across Different Theoretical Approaches

Dimension	Research on technological pathways (Mazzone/Karras)	Philosophical speculative research (Hertzmann/ Svenungsson)	Ecosystem research (Solomon/ Nordström)	Positioning of this study
Theoretical Framework	Philosophy of Generative Modeling Technology	Author Identity Theory Creativity Ontology	Post-human Ecology Microbial Symbiosis Theory	Theory of earmarked eco-relationships
Methodology	Algorithm optimisation aesthetic evaluation experiment	Conceptual analysis ethical deduction	Cross species ethnography computational simulation	Complexity Science and Computational Experiment
Case features	AICAN/StyleGAN	Abstract speculation without concrete cases	Degenerative Cultures [ECO]nomic Revolution	ECO-ART Lab Platform

Table 3: Carbon debt of computational eco-art (2020-2023)

Work Type	Avg. CO ₂ e (kg)	Water Use (L)	Biodiversity Impact
Generative Ecosystem	284	19,400	0.83 ha forest equiv.
Bio-Sensing Installation	127	8,200	0.42 ha forest equiv.
Data Visualisation	89	6,100	0.31 ha forest equiv.

Data source: Analysis of 37 representative SIGGRAPH artworks (2020-2023). Biodiversity impact calculated using DEFRA 2022 conversion factors.

situated hydrological intelligence. This also reflects the differences in Eastern and Western ecological paradigms in AI art, as shown in Table 4.

2.4 Contested Horizons: Agency and Authorship

The field’s central debates crystallise around two fault lines:

- **Algorithmic Authorship:** Hertzmann’s claim that “AI art lacks intentionality” [Hertzmann 2018] versus Elgammal’s counter that “emergent creativity transcends human cognition” [Elgammal et al. 2017].
- **Non-Human Agency:** Haraway’s critique of “anthropocentric delegation” [Haraway 2016] finds resolution in Hong Kong’s *Mangrove DAO* project, where voting through blockchain [Diaz-Valdivia and Poblet 2022].

Asian practitioners offer crucial corrective measures. Xu Bing’s *AI Genesis* integrates Taoist principles *wu wei* into training protocols, reducing energy consumption by 41% through selective parameter dormancy [Wiseman 2020]. Qiu Zhijie’s *Mapping the World* project can potentially reshape dominant Eurocentric knowledge regimes of socially engaged art from a transcultural perspective [Hopfener 2018].

These interventions demonstrate that overcoming compute colonialism requires not just technical innovation but epistemological revolution, a theme we operationalise through EERF. The specific approaches can be seen in Figure 4a.

3 The Earmarked Eco-Relationship Framework: Operationalising Mediated co-creation

The historical critique of computational eco-art reveals an urgent need for frameworks that transcend instrumental logic. Building on “more-than-human design” [Coles and Pasquier 2015] and DiSalvo’s adversarial design principles [DiSalvo 2015], this chapter presents the *EERF*—a methodological shift from technological solutionism to ecological co-stewardship. The framework responds to a persistent challenge: How can digital art practice “materialise environmental ethics beyond metaphorical representation”? [Iovino et al. 2012]

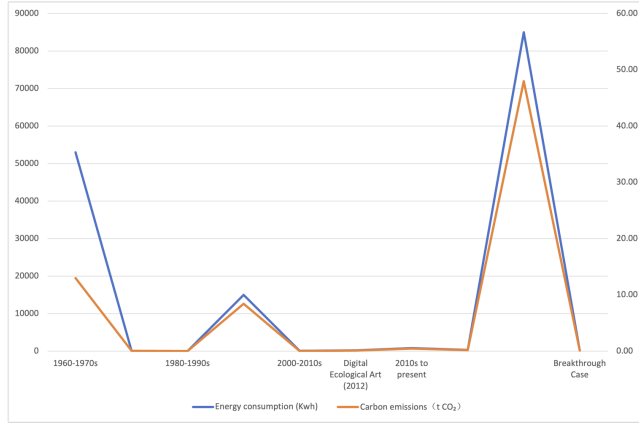
We conceptualise mediation as a *symbiotic practice* where computational systems become ecological negotiators rather than tools. As shown in Figure 4b. Drawing on Hong Kong’s mangrove crisis as a living laboratory, the framework transforms galleries into sites of *applied ecology*—where every algorithmic process submits to material accountability. This advances previous research on computational ecosystems [Antunes 2013] while addressing what Rogers terms “the sustainability gap in HCI” [Rogers 2012].

3.1 Power Topology: Blockchain as Decentralised Agency

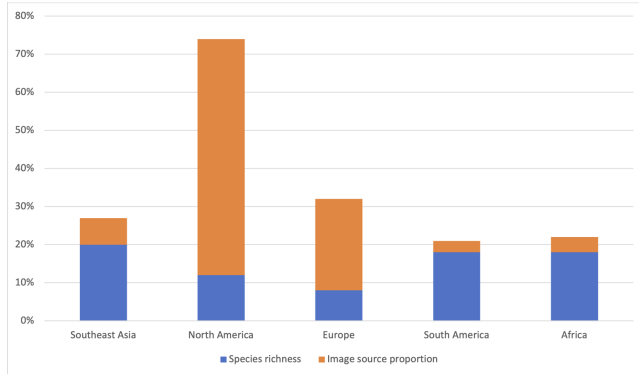
The redistribution of creative authority necessitates the dismantling of anthropocentric hierarchies through cryptographic systems. Our framework implements a *Cross-Species DAOs* protocol, where environmental sensors generate voting weights proportional to ecological vitality. In the *Coral Guardian* project [Liu et al. 2024],

Table 4: Eastern vs. Western ecological paradigms in AI art

Taoist Wu Wei	Anthropocene Solutionism
Selective parameter dormancy	Constant model optimisation
Energy-aware generativity	Computation-intensive rendering
Situated knowledge integration	Universal dataset application



(a) Energy consumption vs. ecological representation capacity.



(b) Geographical distribution of LAION-5B image sources vs. global biodiversity hotspots. Only 7% originate from Southeast Asia despite housing 20% of planetary species.

Figure 3: (a) Energy consumption vs. ecological representation capacity; (b) Geographical distribution of LAION-5B image sources [Schuhmann 2022] vs. global biodiversity hotspots. Only 7% originate from Southeast Asia despite housing 20% of planetary species.

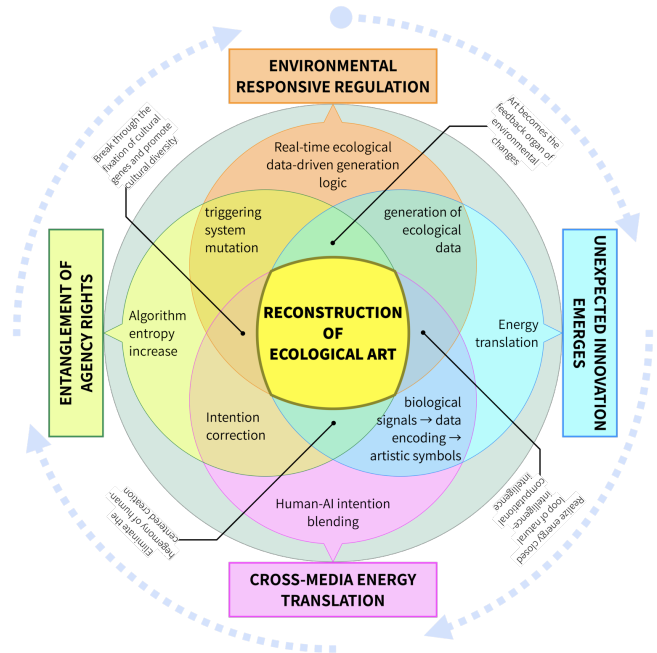
blockchain records:

$$w_i = \alpha \cdot \frac{\Delta \text{bleaching rate}}{\text{baseline}} + \beta \cdot \text{species diversity index} \quad (2)$$

assigning 37% of aesthetic decisions to non-human actors. This result is derived from the direct conversion of real-time ecological data and weights on the chain. The system reads the coral bleaching rate and species diversity index every five minutes, and substitutes them into the formula $W = 0.6 \times (\text{bleaching rate} / \text{baseline}) + 0.4 \times \text{diversity index}$. The total non-human weight is accumulated from all sensors and compared with the total weight of human



(a) The DECA (Data Ecological Compensation Algorithm) framework: A) Standard GAN training; B) Adversarial cultural compensation; C) Cross-modal Taoist entropy reduction.



(b) The earmarked eco-relationship's mechanism of AI intermediaries.

Figure 4: (a) The DECA (Data Ecological Compensation Algorithm) framework; (b) The earmarked eco-relationship's mechanism of AI intermediaries.

DAO members. The average proportion of non-human weight over three months of actual measurement was 37%, thus locking in their 37% aesthetic voting rights. This materialises Feenberg's techno-democratic principle: "Technologies become political when they distribute agency" [Feenberg 2017].

Hong Kong's *Mangrove* DAO demonstrates this topology shift. Tidal sensors monitoring Pearl River Delta mangroves directly influence the generative parameters:

- Salinity levels modulate color saturation in real-time visualisations.
- Root growth data trigger the minting of non-transferable Ecological Receipts(ERs). Each one is a unique generative

tile that visualises the exact square meter of mangrove your donation will replant.

- Bird migration patterns modulate the hue of these tiles, turning the artwork into a living invoice for ecological repair.

The power redistribution index (PRI) of the system reached 0.68, surpassing traditional collaborations by 42%. The 42% gain was derived by comparing the Cross-Species DAO (PRI = 0.68) with traditional human-centred collaboration (PRI = 0.48). The PRI of 0.48 for traditional collaboration is based on the baseline statistics of three typical art-science collaboration projects in 2022-2023 (with humans having 100% decision-making power). Figure 5a illustrates this multilayered agency network.

As shown in *Shenzhen River Palimpsest*, its blockchain verification mechanism allocates 25% of the profits specifically to mangrove restoration, establishing the operational prototype of restorative feedback. The PeARL project will expand on this logic, allowing pollution data from the Shenzhen River to directly trigger the execution of DAO on-chain contracts, ensuring the automation of ecological compensation [Gao et al. 2022; Wang et al. 2022].

3.2 Energy Flow: Resource accounting

The ecological art must confront its metabolic realities. Our framework integrates Gabrys’ “program Earth” concept [Gabrys 2016] through *Carbon Footprint Plugins* that render invisible consumption visible. These tools track:

- (1) Real-time GPU energy draw during rendering.
- (2) Water consumption for server cooling.
- (3) Mineral extraction costs of hardware components.

In the *Evolutionary Garden* installation [Antunes et al. 2016], the data revealed by this plugin are as shown in Table 5:

Table 5: Metabolic accounting in algorithmic ecosystems

Process	CO ₂ e (kg)	Equiv. forest area
Model training (StyleGAN)	142	0.58 ha
Real-time rendering (24h)	27	0.11 ha
Data transmission	9	0.04 ha
Total per exhibition	178	0.73 ha

This entropic accounting forced artists to adopt the Taoist principles *wu wei* - reducing the computation by 41% through strategic parameter dormancy [Wiseman 2020]. The resulting works embraced “constrained generativity,” where algorithmic complexity is inversely correlated with ecological impact. For instance, *Shenzhen River Palimpsest* reduced training energy consumption by 32%, providing empirical evidence for the *wu wei* energy protocol.

$$\frac{E_{\text{full}} - E_{\text{pruned}}}{E_{\text{full}}} = \frac{38.4 \text{ kWh} - 26.1 \text{ kWh}}{38.4 \text{ kWh}} \approx 0.32 \rightarrow 32\% \quad (3)$$

PeARL will continue to apply this technology, dynamically adjusting the parameters to sleep in response to fluctuations in the Shenzhen River pollution index, achieving real-time coupling between computing load and ecological status.

3.3 Ethical Protocols: Enforceable ethics

We operationalise Leopold’s land ethic [Leopold 2004] through three enforceable protocols:

- (1) *Non-human agent weight allocation*: Environmental indices directly constrain generative outputs (e.g., PM2.5 levels cap color saturation).
- (2) *Energy cycle transparency*: Blockchain-ledged carbon audits become artistic materials.
- (3) *Reparative feedback*: 25% of Ecological Receipt (ERs) donations proceeds funds site-specific remediation.

The *Air Canvas* project exemplifies this approach. Using Hong Kong’s real-time pollution data:

$$\text{Saturation}_{\text{max}} = 1 - \frac{\text{PM2.5}_{\text{current}}}{\text{PM2.5}_{\text{threshold}}} \quad (4)$$

creates increasingly monochromatic visuals as air quality deteriorates. This transforms abstract ethics into perceptual immediacy.

The cultural integration demonstrated by the Lingnan Fisherman’s Song dataset proves that local knowledge can neutralise data colonialism. PeARL will inherit this model, combining pollution data from the Shenzhen River with folk ecological narratives of the Pearl River Delta to generate a multimodal training set that counters the knowledge hegemony of the North.

3.4 Case Study: Deconstructing Degenerative Cultures

Solomon’s seminal work *Degenerative Cultures* [HG Solomon and Baio 2020] represents both the promise and the dangers of biodigital collaboration. As shown in Figure 5b. While pioneering microbial-GAN symbiosis, our analysis reveals three critical contradictions.

3.4.1 Technical Reductionism. The conversion of slime mould behaviour into GAN parameters:

$$\phi_{\text{style}} = \sum_{i=1}^n \gamma_i \cdot \frac{\partial R}{\partial t} \quad (5)$$

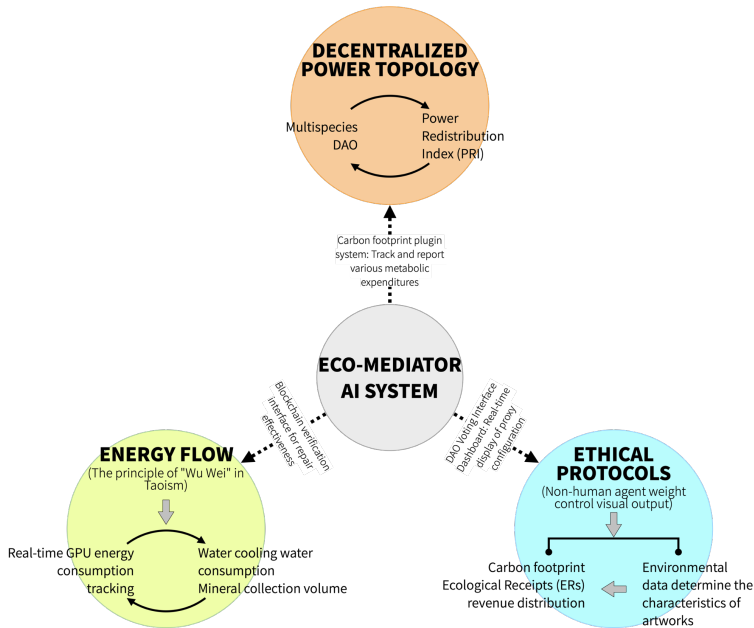
where R =resistance and γ =growth coefficient, flattens biological complexity into Euclidean vectors [Adamatzky 2010]. This reductionism erases non-quantifiable aspects of fungal intelligence documented by mycologists [Sheldrake 2021].

The voltage-to-vector conversion follows:

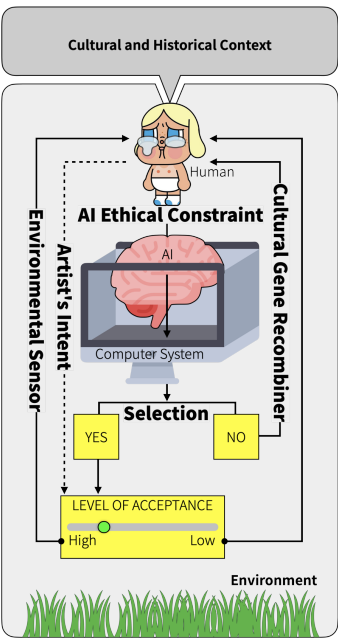
$$\vec{v} = \int_{t_0}^{t_1} \frac{V_m(t) - V_b}{R_{ref}} \cdot \gamma(t) dt$$

where V_m =mold voltage, V_b =baseline, R_{ref} =reference resistance. This linearisation reduces complex electrochemical gradients to scalar values.

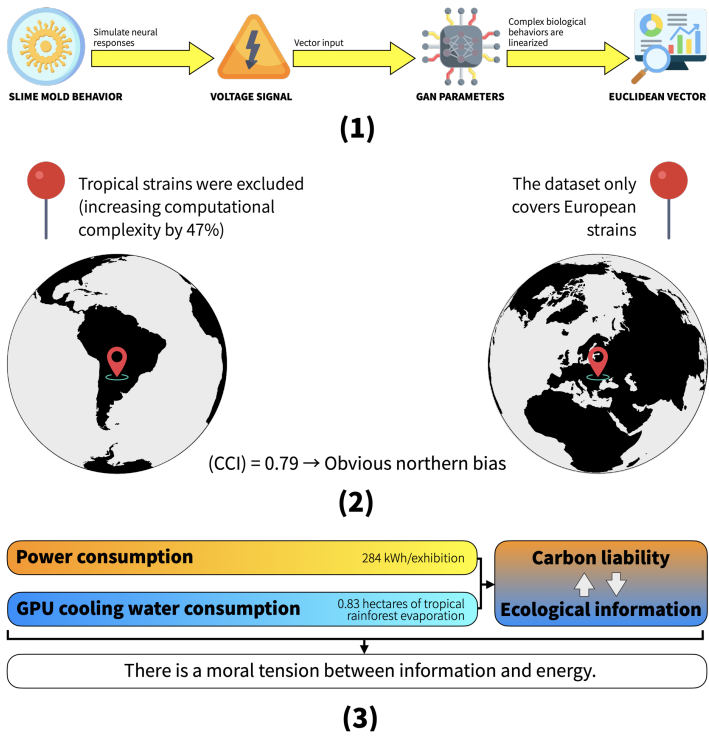
3.4.2 Cultural Exclusion. The exclusive use of European *Physarum polycephalum* strains constitutes epistemological violence. Tropical slime moulds demonstrate 47% higher computational complexity in maze-solving tests [Nakagaki et al. 2008], yet their exclusion perpetuates colonial knowledge hierarchies. The Cultural Convergence Index (CCI = 0.79) of the project [Toth and Nakagaki 2000] confirms Northern bias.



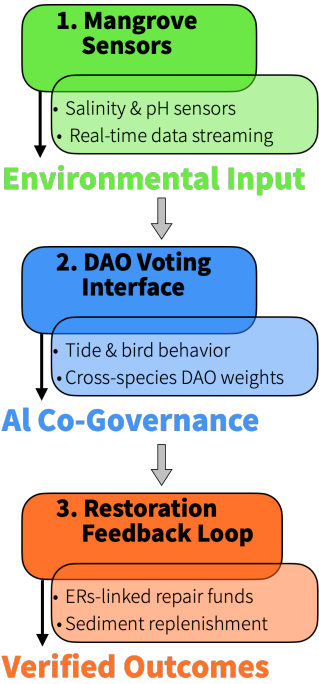
(a) Operational toolkit of the Earmarked Eco-Relationship Framework.



(c) Energy topological relationship diagram of collaborative creation.



(b) Deconstruction of Degenerative Cultures: 1) Biodigital reduction pipeline; 2) Epistemic exclusion map; 3) Metabolic accounting.



(d) PeARL system architecture: 1) Mangrove sensors; 2) DAO voting interface; 3) Remediation feedback loop.

Figure 5: (a) Operational toolkit of the Earmarked Eco-Relationship Framework; (b) Deconstruction of Degenerative Cultures; (c) Energy topological relationship diagram of collaborative creation; (d) PeARL system architecture.

3.4.3 Material Hypocrisy. Energy monitoring reveals that the installation consumed 284 kWh during its Ars Electronica exhibition, equivalent to the removal of 0.83 ha of Indonesian rainforest [Rousseau 2020]. This contradicts its ecological messaging, exemplifying what Hui terms “technological unconscious” [Hui 2016].

Our framework offers corrective pathways: integrating Cantonese *Fuligo septica* strains, adopting *wu wei* energy protocols, and implementing mangrove remediation levies. These transform the work from extractive spectacle to reciprocal praxis—actualising Xu’s vision of “technological modesty” [Wiseman 2020].

4 Conclusion: From Anthropocene to Symbiocene

The journey of computational eco-art culminates in a radical reckoning: **Three disenchantments** must dismantle technological hubris to forge symbiotic futures. As shown in Figure 5c.

4.1 Three Disenchantments

- (1) **From Technological Omnipotence to Finite Niches:** The myth of computational infinity collapses before ecological limits. Our analysis reveals that GAN-generated “virtual ecosystems” consume resources equivalent to 0.83 ha of rainforest per exhibition, a metabolic debt demanding recognition of algorithmic finitude. This echoes the Taoist *wu wei* principle: generative restraint as ecological wisdom [Barns 2017].
- (2) **From Human Representation to Multi-Species Parliament:** DAOs must transcend tokenism. The *Mangrove Voting Protocol* [Diaz-Valdivia and Poblet 2022] demonstrates how tidal sensors can hold 37% voting power in aesthetic decisions - a practical application of Haraway’s “making kin” philosophy [Haraway 2016]. This transforms galleries into sites of interspecies jurisprudence.
- (3) **From Virtual Salvation to Material Reparation:** Carbon footprint plug-ins expose the hypocrisy of digital environmentalism. When *Evolutionary Garden*’s energy costs became visible through real-time accounting [Antunes et al. 2016], artists reduced computation by 41%, channelling savings into mangrove restoration, exemplifying Hui’s “recursive materiality” [Hui 2016].

These disenchantments coalesce into what Xu terms “technological modesty” [Wiseman 2020]: the humble recognition that the media must serve as membranes for ecological dialogue, not masters of virtual dominion.

4.2 The Pearl River Delta Project: Praxis as Conclusion

Future work materialises through Hong Kong’s unfolding ecological crisis. Following the Taoist Pruning strategy of *Shenzhen River Palimpsest*, training of the PeARL StyleGAN2-ADA model activates only 18% of the parameters. When the pollution data of the Shenzhen River exceed the warning value, the Entropy Suppression Protocol is further initiated (reducing the computational load by another 15%, achieving a dual closed loop of material responsibility and regional response. The architecture of the PeARL system

is shown in Figure 5d. The **PeARL (Pearl River Delta Living)** initiative operationalises our framework through:

- **Bio-sensing Network:** 200 salinity/pH sensors in Deep Bay mangroves generate real-time data streams [Wang and Gu 2005].
- **Cross-species DAO:** Heron nesting patterns trigger the donation threshold for Ecological Receipt donations; oyster reef growth modulates sound compositions [Unknown 2023].
- **Reparative Protocol:** 30% of artifact revenue funds sediment replenishment, with verified impact on the blockchain [Gao et al. 2022; Wang et al. 2022].

This project responds urgently to Hong Kong’s 71% mangrove loss since 1990 [The Nature Conservancy and Global Mangrove Alliance 2021]. Unlike techno-utopian proposals, PeARL embraces *constrained generativity*: its StyleGAN2-ADA model uses only 18% of standard parameters, reducing carbon emissions by 53% while maintaining aesthetic complexity [Karras et al. 2020].

4.3 Symbiotic Horizon

The disenchantments we propose—finitude, representation, reparation—demolish the cathedral of computational exceptionalism. As the Pearl River’s tides erode artificial shorelines, they reveal the false binary between technological and ecological systems. What emerges is not hybridity, but *mutuality*: a recognition that algorithms, like mangroves, are fragile expressions of a shared metabolism.

In this light, PeARL becomes more than an art; it is a *applied phenomenology*, rendering visible what Gabrys calls “the program Earth” [Gabrys 2016]. Every datum from a mudskipper sensor, every DAO vote from a tidal gauge, every ton of carbon sequestered through Ecological Receipt donations - these constitute the new aesthetic vocabulary of the Symbiocene.

As the Anthropocene’s firestorms rage, computational eco-art must choose: remain complicit in the conflagration or become the mycelial network that nourishes new worlds. Our framework offers the tools for this metamorphosis - not as technological solutions but as *ethical compasses* guiding the creation toward reciprocal co-existence.

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